5.6 Product specifications

The technical architecture of open.lens is designed for maximum accessibility, utilizing a combination of affordable **off-the-shelf parts** and **free open-source files**. The device is fully customizable, allowing the end-user to modify and print components according to specific needs. The hardware ecosystem is divided into three main categories:

Structural components

These parts provide a robust, low-cost frame that ensures precision and modularity, allowing modules to be easily positioned along the axes.

- 30x30 alluminum profiles
- M8 thumb screws
- M8 sliding nuts for 30x30 aluminum profiles

Open-source components

These elements are designed to be manufactured locally by the user, with geometries optimized for additive manufacturing to minimize waste and support structures while encouraging design personalization.

- Clamp
- Joints
- Camera module
- Accessories

Electrical and optical components

By repurposing budget-friendly webcams and using widely available or second-hand photographic lenses, the project eliminates the need for expensive proprietary hardware, making high-quality magnification accessible through easily reachable or household items.

- Webcam
- Camera lens
- Ring light
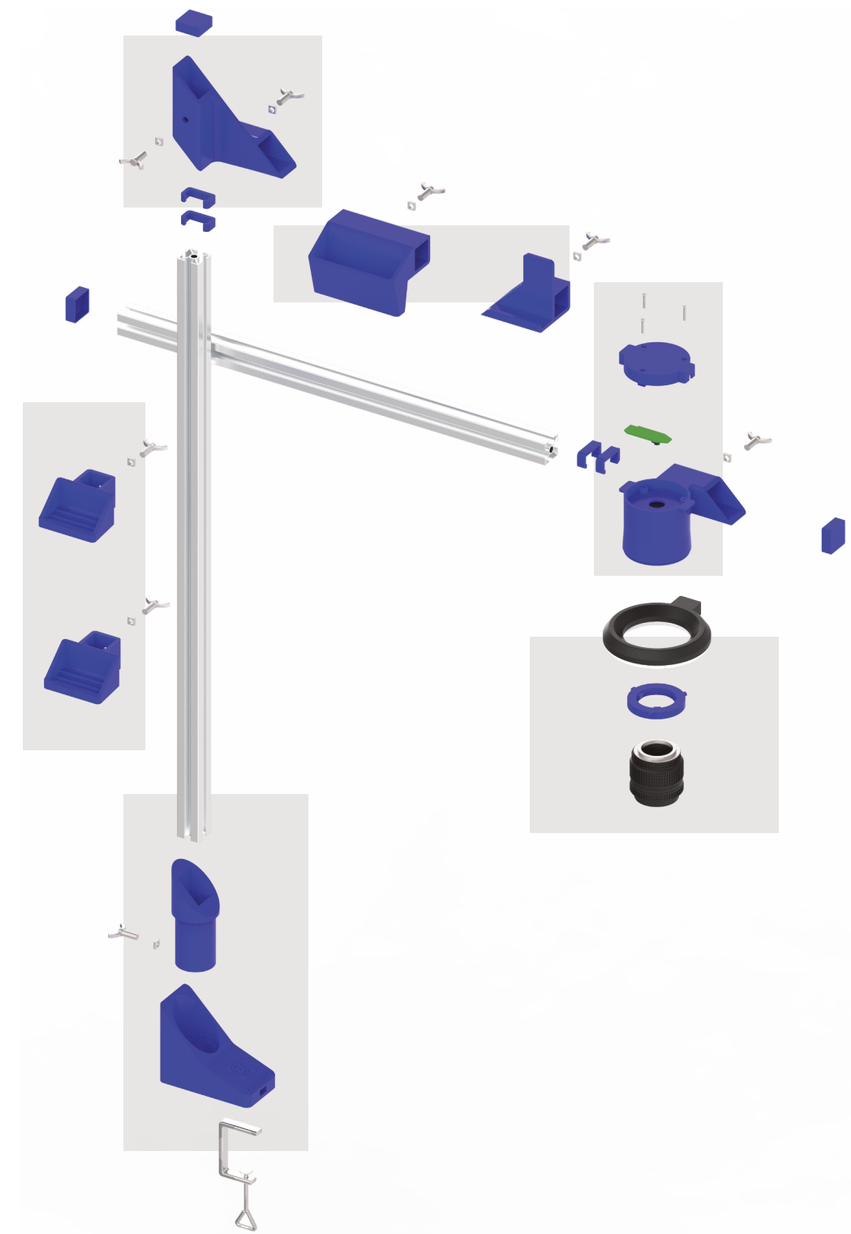
Fig. 112
Exploded view of open.lens crane mode